

West Hartford Deck — Lumber Yard

Purchase List

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Scope: 23'-6" × 16'-3" attached pressure-treated deck with MoistureShield Vision composite decking — all lumber, composite, connectors, fasteners, flashing, and moisture protection items purchasable at a full-service lumber yard.
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1. Shims & Bearing Hardware

Item	Spec	Qty	Notes
1/4" cedar or HDPE rot-resistant shims	Min. 1.5" × 3.5" × 1/4" shim stock	40	2 shims per saddle × 2 sides × 10 saddles = 40 total. Use cedar or HDPE only — no OSB, no wood composite, no plywood.
1/2" HDG lag bolt (girder-to-saddle)	1/2" × 3" HDG	20	2 per saddle × 10 saddles, up into girder bottom.

2. Lumber — Pressure-Treated SYP, Ground-Contact (UC4B)

All PT lumber: Southern Yellow Pine, ground-contact rated (UC4B), KDAT (kiln-dried after treatment) preferred. Verify net dressed length at the yard before delivery.

2a. Ledger

Item	Size	Length	Qty	Notes
Ledger board	2×10 PT UC4B	20'-0"	1	

2b. Girders (Doubled 2×10 PT)

Each girder line is built from two plies using a staggered-splice pattern. Ply joints must land over a pier bearing point — never in mid-span.

Ply	Pieces	Cut lengths	Joint locations	Source stock
Ply 1	2 pieces	11'-6" + 11'-6"	Butt joint over center pier at x=11'-6"	2 × 12-ft 2×10 PT (6" waste each; drops usable for blocking)
Ply 2	3 pieces	5'-9" + 11'-6" + 5'-9"	Joints over piers at x=5'-9" and x=17'-3"	1 × 12-ft 2×10 PT (center piece) + 2 × 8-ft 2×10 PT (end pieces, 2¼" waste each)

Board count per girder line (2 plies): - Ply 1: 2 × 12-ft boards - Ply 2: 1 × 12-ft board + 2 × 8-ft boards - **Total per girder line: 3 × 12-ft + 2 × 8-ft**

Order for 2 girder lines with 1 spare board each length:

Item	Size	Length	Qty	Notes
Girder ply — 12-ft piece	2×10 PT UC4B	12-ft	7	2 girder lines × 3 pieces per line (2 for Ply 1 + 1 for Ply 2 center) = 6 needed + 1 spare. Field-cut Ply 1 pieces to 11'-6" (6" waste per board — save drops for blocking).
Girder ply — 8-ft piece	2×10 PT UC4B	8-ft	5	2 girder lines × 2 pieces per line (Ply 2 end pieces, each cut to 5'-9" with 2¼" waste) = 4 needed + 1 spare.

2c. Joists (Face-Hung, Two Segments Per Row)

Item	Size	Length	Qty	Notes
Joist segments (regular)	2×10 PT UC4B	~7'-11¼" clear span each (order 8-ft)	48	24 regular rows × 2 segments each. Each segment spans ~7'-11¼" clear

Item	Size	Length	Qty	Notes
rows, both halves)				(ledger face to mid-girder face for inboard segment; mid-girder far face to outer-rim girder face for outboard segment). Order 8-ft, field-cut to fit.

2d. Blocking

Blocking runs x-direction (perpendicular to joists). Two blocking rows recommended to stiffen mid-span.

Item	Size	Length	Qty	Notes
Field blocking (lateral stiffness)	2×10 PT UC4B	~10¼" (joist bay width at 12" o.c.)	~46	Two blocking rows at y=4'-0" (mid-span of inboard bay) and y=12'-0" (mid-span of outboard bay). Each row spans 23'-0" with one block per joist bay. 23 joist bays × 2 rows = 46 blocks. Cut from 12-ft or 8-ft drops from joist material.
House-side PF backing blocks	2×10 PT UC4B	~9" each	~25	Short pieces at 12" o.c. between ledger face and first joist face; back the house-side picture-frame board.

Tip: Field blocking cuts from joist drop-offs (8-ft stock yields one segment + 4-ft drop → 4 blocking cuts). Order 4 extra 8-ft 2×10s beyond joist count if using dedicated blocking stock, or source from drops.

2e. Lumber Order Summary

Category	Stock Size	Qty to Order
Ledger	2×10 PT 20-ft	1
Girder plies	2×10 PT 12-ft	7 (6 needed + 1 spare)
Girder plies	2×10 PT 8-ft	5 (4 needed + 1 spare)
Joists	2×10 PT 8-ft	55 (50 needed + ~10% waste)
Blocking (field + house-side)	2×10 PT — cut from drops	— (or order 4 extra 8-ft boards for dedicated blocking stock)

3. Decking, Picture Frame & Fascia

All MoistureShield Vision composite, capped, DiamondDefense scratch coat, 50-yr fade & stain warranty. Available at Home Depot, Lowe's, Richards Building Supply (Hartford metro).

3a. Field Planks

Field decking runs x-direction (parallel to house) in two halves of equal width, each half spanning 11'-9" from the left or right PF inboard edge to the center breaker board.

Item	Brand / Product	Width	Thickness	Cut Length	Qty	Notes
Field decking planks — left half	MoistureShield Vision	5.4"	1.0"	11'-9" (from 12-ft stock)	32	16 plank rows left of center × 1 piece each; butts at center divider breaker.
Field decking planks — right half	MoistureShield Vision	5.4"	1.0"	11'-9" (from 12-ft stock)	32	16 plank rows right of center × 1 piece each; butts at center

Item	Brand / Product	Width	Thickness	Cut Length	Qty	Notes
Field planks (ordered)	MoistureShield Vision 12-ft	—	—	12-ft boards, field-cut to 11'-9"	72	divider breaker. 64 pieces needed + 10% waste = 70.4 → order 72 × 12-ft Vision boards.

Stock option note: 20-ft Vision boards can be cut to two 11'-9" pieces with only 6" waste per board, yielding 36 × 20-ft instead of 72 × 12-ft. Confirm 12-ft availability at Richards Building Supply (West Hartford) before ordering. If 12-ft is in stock, prefer it.

3b. Picture-Frame Boards (4 Sides plus Center Divider)

Side	Length	Ordered As	Qty	Corner Treatment
House-side PF	23'-6"	24-ft board, field-cut; or 2× 12-ft spliced over a joist	1	Butt at left/right against house wall
Far-side PF	23'-6"	Same	1	45° mitered at left/right outside corners
Left-side PF	16'-3"	18-ft board field-cut to 16'-3"; or splice 2 shorter boards over a joist	1	Mitered at far corner; butt at house end
Right-side PF	16'-3"	Same	1	Mitered at far corner; butt at house end
Center divider breaker board	16'-0"	Single 16-ft or 20-ft board field-cut to 16'-0"	1	Face-screwed into center joist top; no miter; butts house-side PF on one end and far-side PF on the other

Center divider detail: The breaker board sits at x=11'-9" finish position. The center framing joist beneath it is at x=11'-6" framing centerline. The breaker is face-screwed to the center joist top with Cortex screws + Vision plugs at approximately 12" o.c. — 16 screws + 16 plugs for the 16'-0" run.

3c. Fascia Boards (3 Exposed Sides)

MoistureShield Vision composite fascia: 0.67" thick, 11.25" tall.

Side	Length	Ordered As	Qty	Notes
Far-side fascia	23'-0"	24-ft board field-cut; or 2× 12-ft	1	Full run; backs far-side PF.
Left-side fascia	16'-0"	16-ft board, use full length	1	Full run on left (stair) side.
Right-side fascia	16'-0"	16-ft board, use full length	1	Full run on right side.

3d. Decking Board Order Summary

Product	Length	Qty to Order
MoistureShield Vision field planks	12-ft	72
MoistureShield Vision center divider breaker	16-ft or 20-ft (1 piece)	1
MoistureShield Vision PF boards	24-ft (house-side + far-side)	2
MoistureShield Vision PF boards	18-ft (left-side + right-side, 16'-3")	2
MoistureShield Vision fascia	24-ft (far-side)	1
MoistureShield Vision fascia	16-ft (left-side + right-side — exact length)	2

4. Connectors & Hangers

All connectors: hot-dip galvanized (HDG), ACQ-compatible. Simpson Strong-Tie specified throughout.

Item	Model	Qty	Notes
Girder end-to- ledger hanger (doubled 2×10)	Simpson HU210-2 (2-ply, HDG)	2	One at each end of each girder line at the ledger (x=0' and x=23'-0") — only the 2 end points of the mid-girder line land at the ledger; outer-rim girder does

Item	Model	Qty	Notes
Joist-to-ledger face-mount hanger	Simpson LUS210Z HDG	25	not connect to ledger. 25 joist rows × 1 hanger each at ledger end.
Joist-to-mid-girder face-mount hanger (house-side face)	Simpson LUS210Z HDG	25	25 joist rows × 1 hanger each hanging from mid-girder house-side face.
Joist-to-mid-girder face-mount hanger (far-side face)	Simpson LUS210Z HDG	25	25 joist rows × 1 hanger each hanging from mid-girder far face.
Joist-to-outer-rim-girder face-mount hanger (inboard face only)	Simpson LUS210Z HDG	25	25 joist rows × 1 hanger each; joists terminate at outer-rim girder inboard face.
Hanger count total — LUS210Z: 25 (ledger) + 25 (mid-girder house face) + 25 (mid-girder far face) + 25 (outer-rim girder) = 100 LUS210Z . Add 5% overage: order 110 LUS210Z (round up to box quantity for convenience).			

5. Fasteners

All fasteners: HDG or stainless (ACQ-compatible). No electroplated zinc.

5a. Ledger-to-House

Item	Spec	Qty	Notes
Ledger lag bolts (field, between hanger locations)	½" × 4" HDG lag	34	½" HDG lag at 18" o.c. two-row staggered between hanger-connection zones.
Through-bolts at girder-to-ledger (2 locations)	½" × 6" HDG carriage bolt + nut + 2 washers	12	6 per cluster × 2 cluster locations (x=0', x=23'-0") at mid-girder line end-to-ledger connections.
Ledger through-bolt backup blocks (interior band joist)	2×10 PT scrap	2 short pieces	Back-blocking behind ledger at each carriage-bolt cluster.

5b. Girder Assembly (Doubled 2-Ply)

Per AWC/DCA-6 prescriptive practice for a 2-ply girder: ½" HDG carriage bolts at 16" o.c. is the clean, code-recognized spec. Through-bolts engage both plies positively and are the straightforward choice for a residential deck girder of this span.

Item	Spec	Qty	Notes
Girder ply-to-ply carriage bolts + nut + washer	½" × 3½" HDG carriage bolt	~52	~1 per 16" o.c. × 46' total girder run (2 lines × 23') ÷ 1.33' spacing = ~35 bolts; double for 2-ply bolt-through = ~70; realistically ~52 with bolt-through counting once per paired hole. Order 52 sets (bolt + nut + 2 washers per set). Verify DCA-6 Table 2 for your specific span and loading.

5c. Joist Hangers

Item	Spec	Qty	Notes
Joist hanger nails/screws — LUS210Z	Simpson N10 (10d × 1½") or SDS ¼"×1½" screws per Simpson spec sheet	~1,400	100 hangers × ~14 nail holes per LUS210Z = 1,400; buy 1,500.
HU210-2 fasteners (girder-end-to-ledger)	Per Simpson spec (SDS ¼"×1½" or N10 as spec sheet requires for 2-ply)	~60	2 hangers × ~30 fastener holes each.

Critical: Verify exact nail/screw specification on the Simpson packaging for each hanger model. Do not substitute nail sizes.

5d. Decking Fasteners

Item	Brand / Spec	Qty	Notes
Hidden fasteners — field planks	Tiger Claw TC-G clips	~700	25 joist rows × 32 planks × ~16 clip points per plank = 512 clips per half × 2 halves; buy

Item	Brand / Spec	Qty	Notes
#8 stainless screws for TC-G clips	Tiger Claw specified (stainless, 1-5/8")	~700	700 with overage. One per clip.
Cortex face-screws + color-matched plugs — PF boards (4 sides + center divider)	Cortex for MoistureShield (must match Vision colorway)	5 boxes (100-ct each)	~4 screws per lineal foot × ~80 lf four-sided PF + 16 lf center divider = ~384 screws.
Stainless trim screws — fascia to rim/drop-board	#10 × 2½" stainless	1 box (100-ct)	~3 screws per lineal foot × ~55 lf of fascia ≈ 165.

5e. Lateral Tie Fasteners

Item	Spec	Qty	Notes
DTT2Z structural screws	SDS ¼"×4½" per Simpson spec	~32	~8 per DTT2Z device × 4 devices.

6. Flashing & Moisture Protection

Item	Brand / Spec	Coverage / Size	Qty	Notes
Peel-and-stick flashing tape (ledger)	Henry Blueskin VP160 or Carlisle CCW-705	6" wide roll	2 rolls (75 lf each)	Apply to house sheathing behind ledger before ledger install; 23'-0" run.
Z-flashing, aluminum	2" leg × 2" leg, 10-ft sticks	3 sticks		Tucked behind siding/housewrap above top of ledger.
Kick-out flashing	Aluminum or plastic, pre-bent	2		At each end of ledger where

Item	Brand / Spec	Coverage / Size	Qty	Notes
				siding terminates.
Joist / girder top tape	Vycor Deck Protector 2" or Trex Protect 2½"	2" wide roll, 50 lf per roll	11 rolls	Apply to top of every joist segment (~400 lf), both girder top runs (~92 lf), and ledger top (~23 lf) = ~515 lf total.
End-grain sealant	Copper Green Wood Preservative or Anchorseal 2	1-qt brush-on	2 qt	Seal all PT field cuts immediately after cutting.
Backer rod + polyurethane sealant	Sikaflex 1A or similar, gray	1 tube	3 tubes	Ledger-to-house seam caulk.
Ledger spacers / standoffs	Starborn Camo LedgerUp composite or PT shims	—	as needed	If siding extends behind ledger location.
Drip cap (head flashing)	Aluminum drip cap, 2" face	24-lf sticks	1 stick	Over ledger at top if housewrap alone is insufficient.

7. Lateral Tie System (IRC R507.9)

Item	Model	Qty	Notes
Deck-to-house lateral connector	Simpson DTT2Z (or MTWDeck equivalent)	4	IRC R507.9 minimum; deck is 16'-3" wide. Each device connects deck framing through ledger to interior floor framing. Basement accessible.

Placement: space DTT2Z devices evenly along ledger, avoiding the HU210-2 hanger cluster zones. Confirm exact placement with structural engineer or West Hartford building official.

Closing Notes

Far-edge step assembly omitted. The far-edge step ribbon (doubled 2×6 PT board running along the far rim) was deliberately removed from this list. That step sits on precast concrete blocks, which are out of scope for a lumber-yard purchase order. The step assembly — ribbon, ribbon lag bolts, and ply-to-ply screws — should be addressed in a separate concrete/masonry materials list once the step foundation is finalized.

Pylex screw piers are not listed here; they are a specialty pier product ordered separately from a lumber yard. Confirm pier count and order with your pier supplier before framing begins.

Seal all PT field cuts with end-grain sealant immediately after cutting — this is the single most skipped step on PT decks and the first place rot starts.